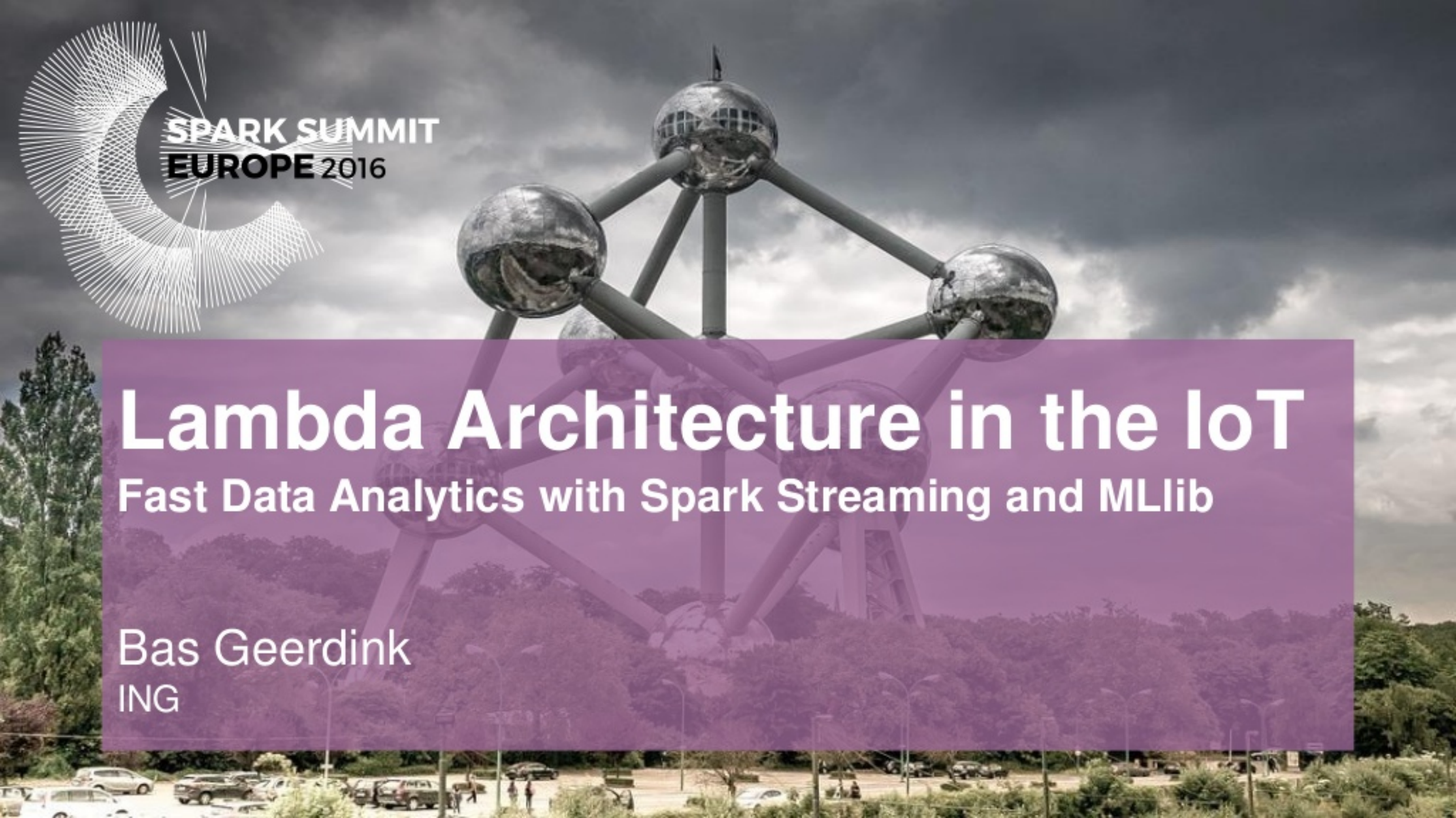


SPARK SUMMIT EUROPE 2016

Lambda Architecture in the IoT

Fast Data Analytics with Spark Streaming and MLlib

Bas Geerdink
ING

Who am I?

Bas Geerdink

- Chapter Lead in Analytics area at ING
- Master degree in Artificial Intelligence and Informatics
- Spark Certified Developer

- @bgeerdink
- https://www.linkedin.com/in/geerdink

The Internet of …

What's new in the IoT?

- Data
  - Streaming data from more sources
- Use cases
  - Combining data streams
- Technology
  - Fast processing and scalability
- Challenges
  - Security: encrypt the sensors/network/server

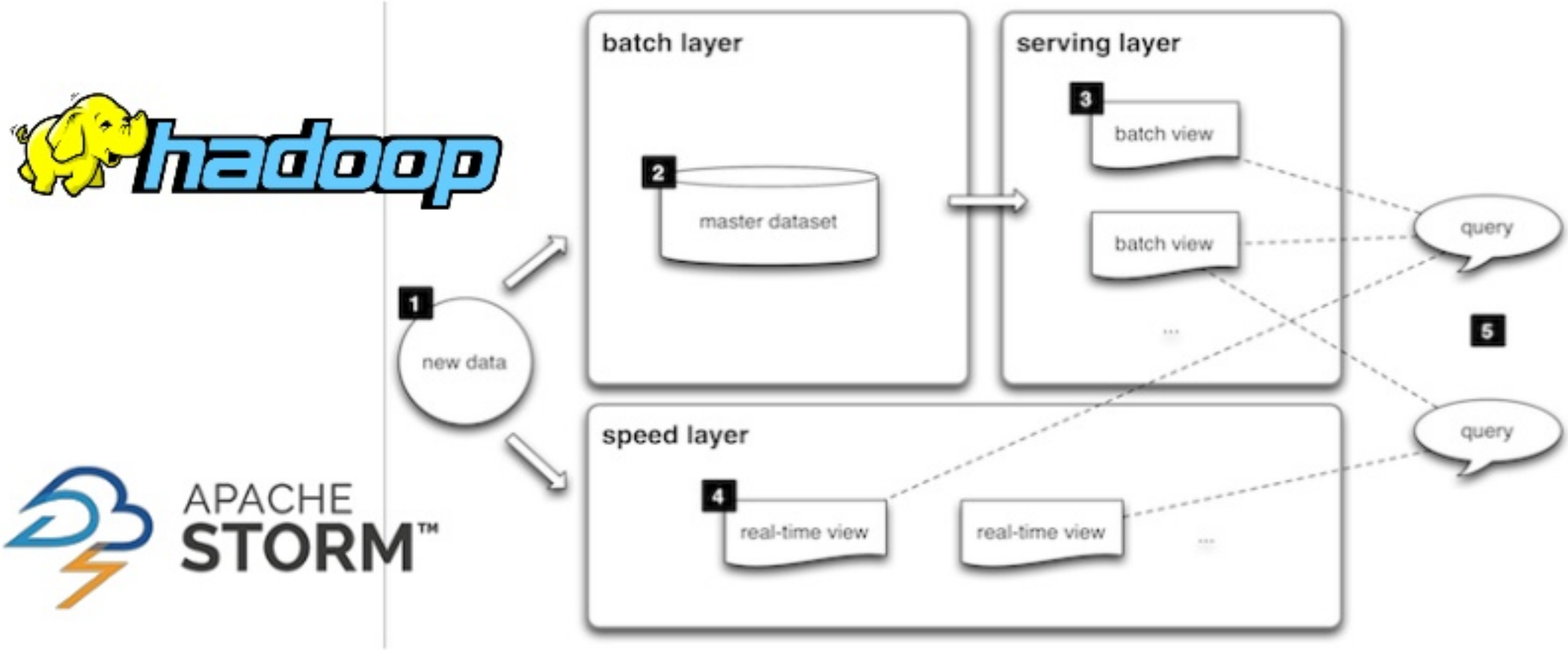
What do we want in the IoT?

- FAST data: process stream of events (sensory data)
- BIG data: process files/tables in batches (static data)

- ONE analytics engine
- ONE querying/visualization tool

- Scalable environment
- Reactive software

Lambda Architecture



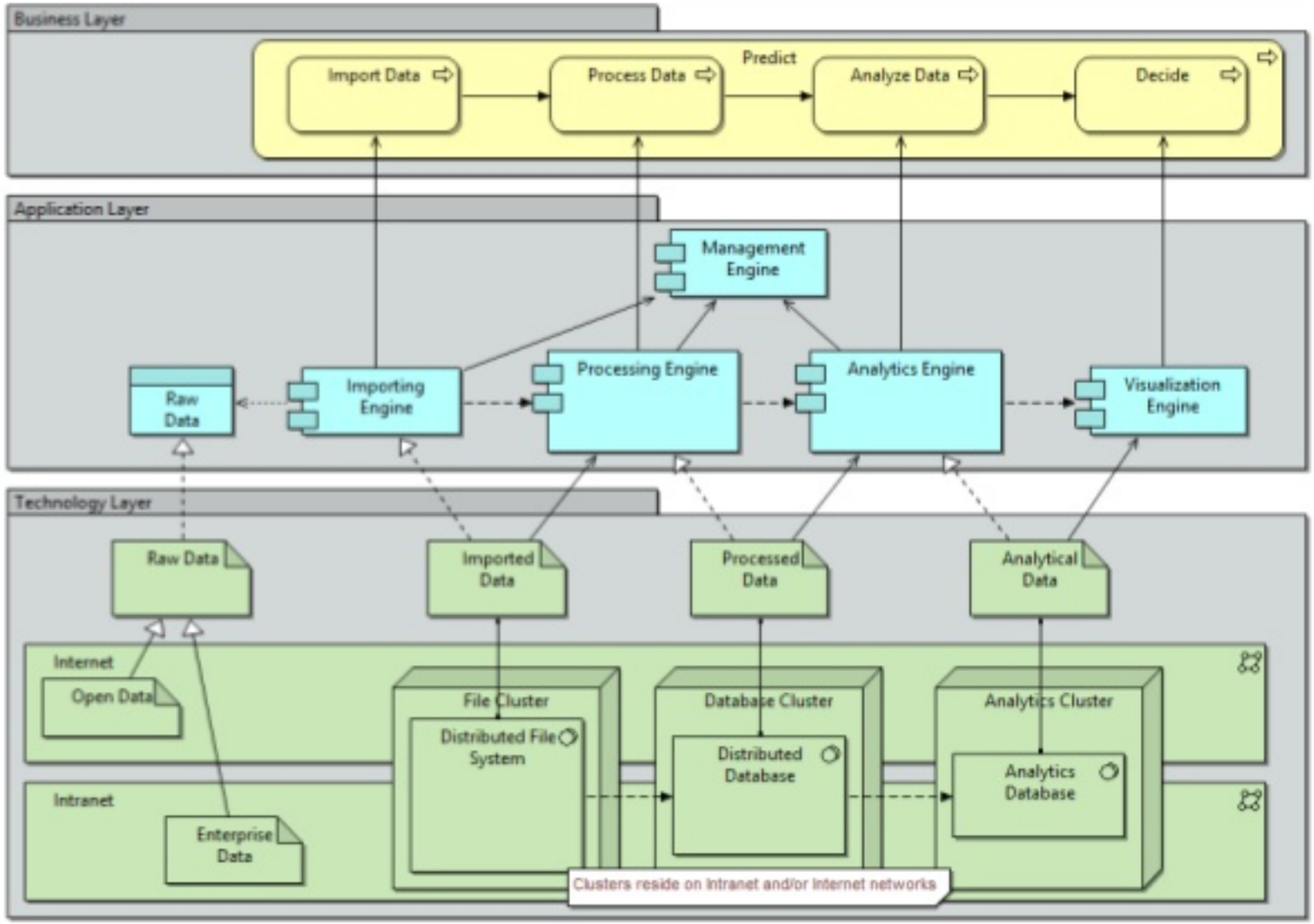
Source: Nathan Marz (2013)

Gamma/Kappa/Omega/… Architecture

- Lambda Criticism:
  - Too complex
  - Two code bases
  - Two data stores

- Alternatives:
  - Treat a stream as a mini-batch
  - Treat a batch as a stream of records
  - Combine batch and stream within one system

Big Data Reference Architecture



Source: Geerdink (2013)

Use case: Smart Parking

Recommend the best car park when driving to Amsterdam

→ Show the car park with the highest score, determined by

- Batch (every x minutes), data from car parks
- Stream (every x seconds), GPS data of cars

Gemeente Amsterdam

Wonen
Werken
Vrije tijd
Buurten

Actueel
Veelgevraagd
Bestuur en organisatie
Alle onderwerpen

Contact
English
Adressengids
Nieuwsbrief

Home > Parkeren & verkeer > Parkeren + Reizen (P+R)

Zoeken...

Parkeren + Reizen (P+R)

Parkeer uw auto voor € 8,00 per 24 uur (na 10.00 uur voor € 1,00 per 24 uur) aan de rand van Amsterdam en reis met openbaar vervoer naar het centrum.

P+R locaties

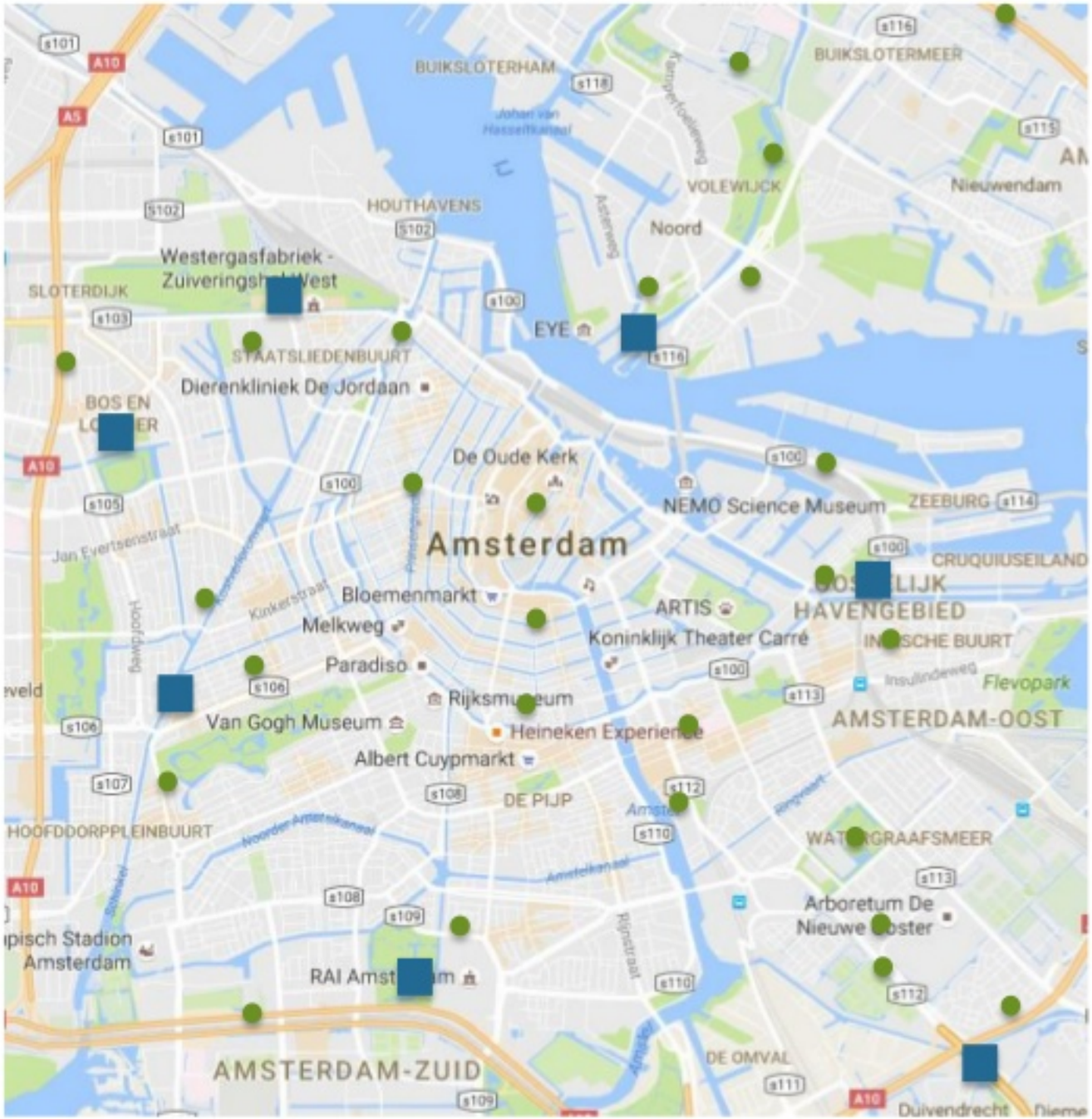
Meer informatie per P+R locatie en uitleg hoe het werkt:

- P+R ArenA
- P+R Zeeburg I en II
- Weekend P+R VUmc
- P+R Bos en Lommer
- P+R RAI
- P+R Sloterdijk
- P+R Olympisch Stadion
- Welke P+R kies ik?

Actuele beschikbaarheid P + R parkeerplekken

Laatste update: 25-okt-2016 00:21 (elke minuut)

P+R locatie	Beschikbaarheid	Parkeerplekken
P+R ArenA	Vrij	1.440
P+R Zeeburg 2	Vrij	322
P+R Olympisch Stadion	Vrij	161
P+R Zeeburg 1	Vrij	60
P+R Sloterdijk	Geen informatie	
P+R Bos en Lommer	Vrij	21


BUIKSLOTERHAM
BUIKSLOTERMEER
VOLEWIJCK
Nieuwendam
HOUTHAVENS
Noord
Westergasfabriek -
SLOTERDIJK
EYE
STAATSLIEDENBUURT
Dierenkliniek De Jordaan
De Oude Kerk
NEMO Science Museum
ZEEBURG
Amsterdam
CRUQUIUSEILAND
Bloemenmarkt
ARTIS
HAVENGEBIED
Melkweg
Koninklijk Theater Carré
Paradiso
Flevopark
Van Gogh Museum
Heineken Experience
AMSTERDAM-OOST
Albert Cuypmarkt
DE PIJP
HOOFDDORPPLEINBUURT
Arboretum De
RAI
AMSTERDAM-ZUID
DE OMVAL
Duivendrecht

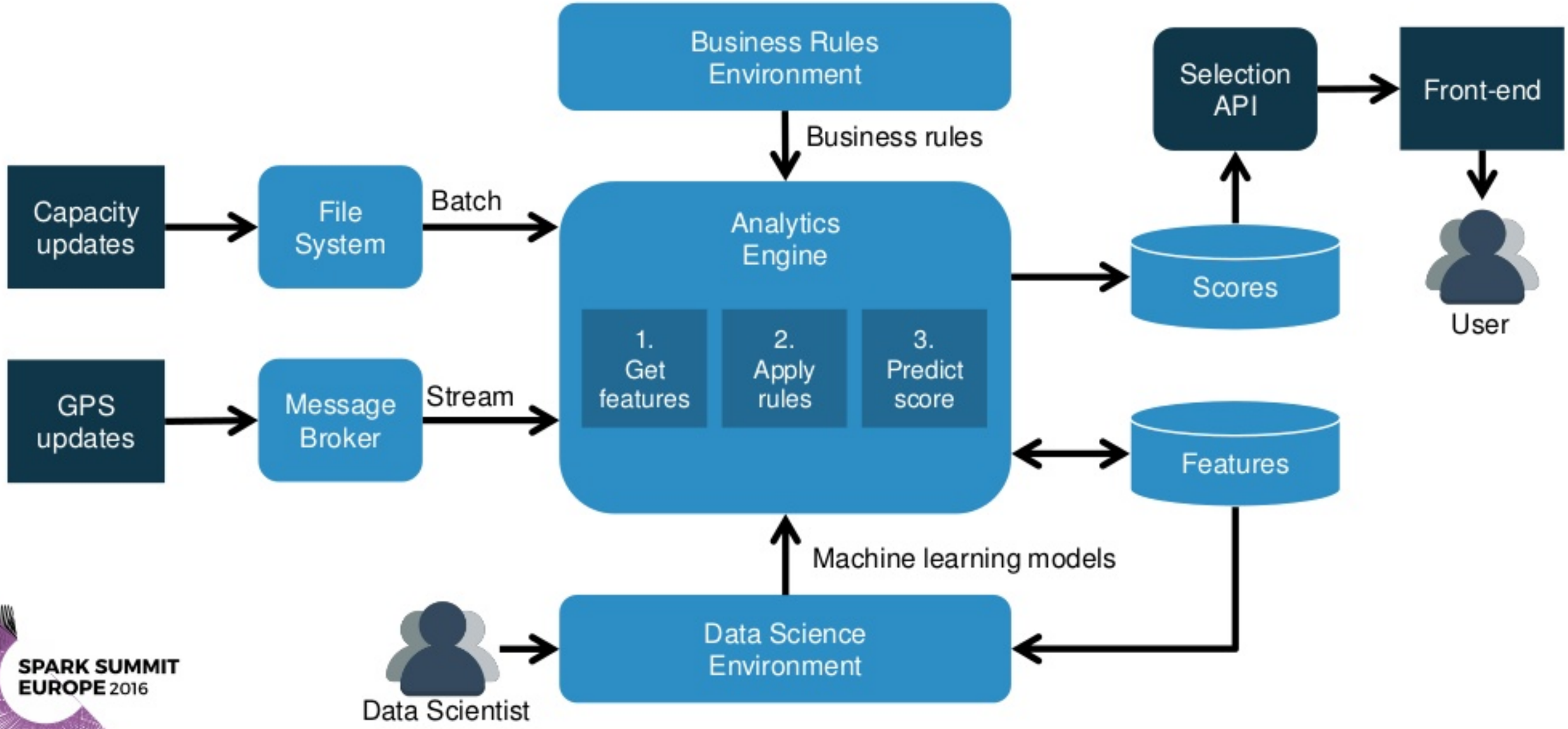
Stream Process

- Get car events (GPS data)
- Filter events (business rules)
- Store events
- For each car park in the neighborhood:
  - Get feature set (location, capacity, usage, …)
  - Combine event with previous events (running sum)
  - Update feature set
  - Predict score (machine learning) and update database

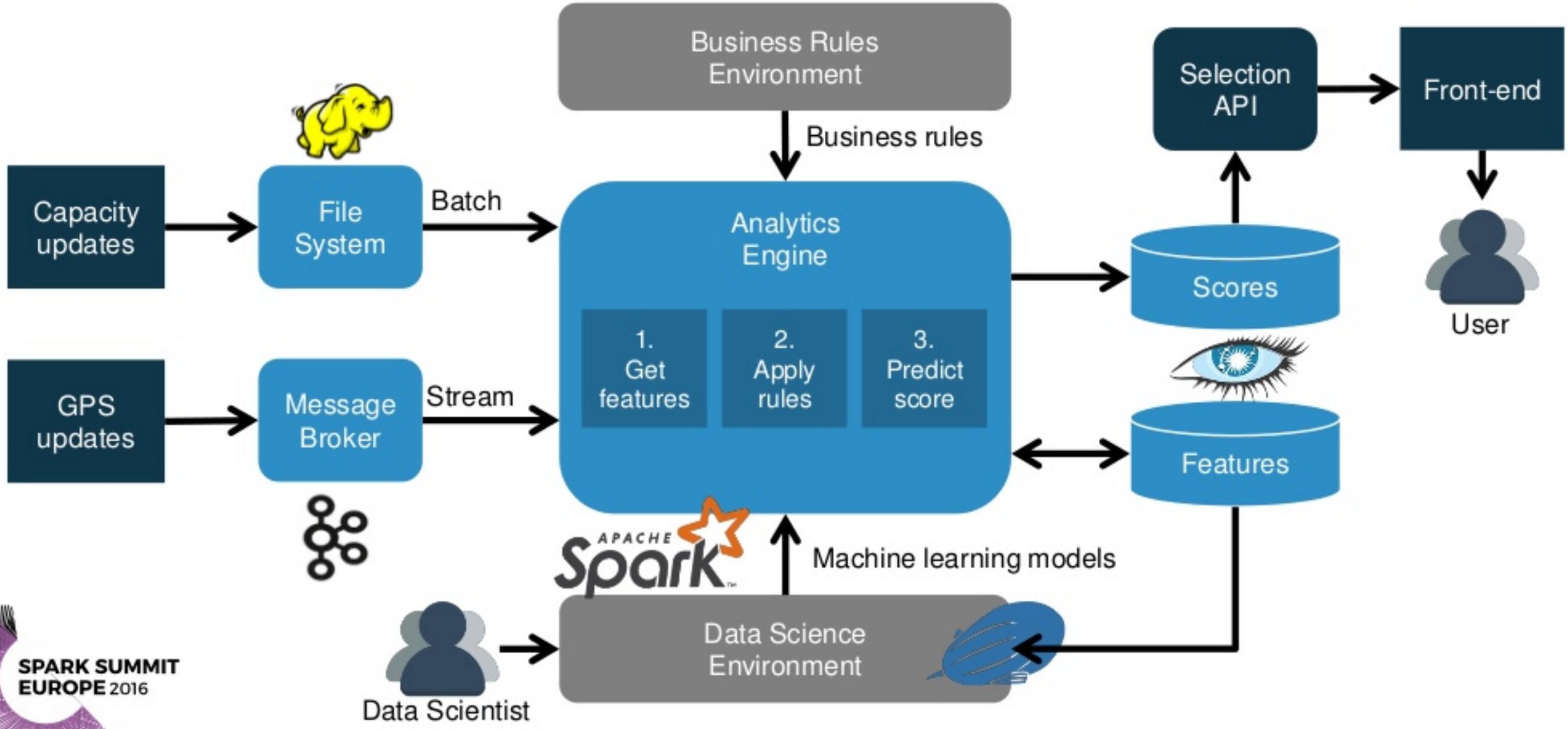
Batch Process

- Get car park data
- Clean up (remove old car data)
- For each car park in the data set:
  - Get recent set of car data (running sum)
  - Update feature set
  - Predict score (machine learning) and update database

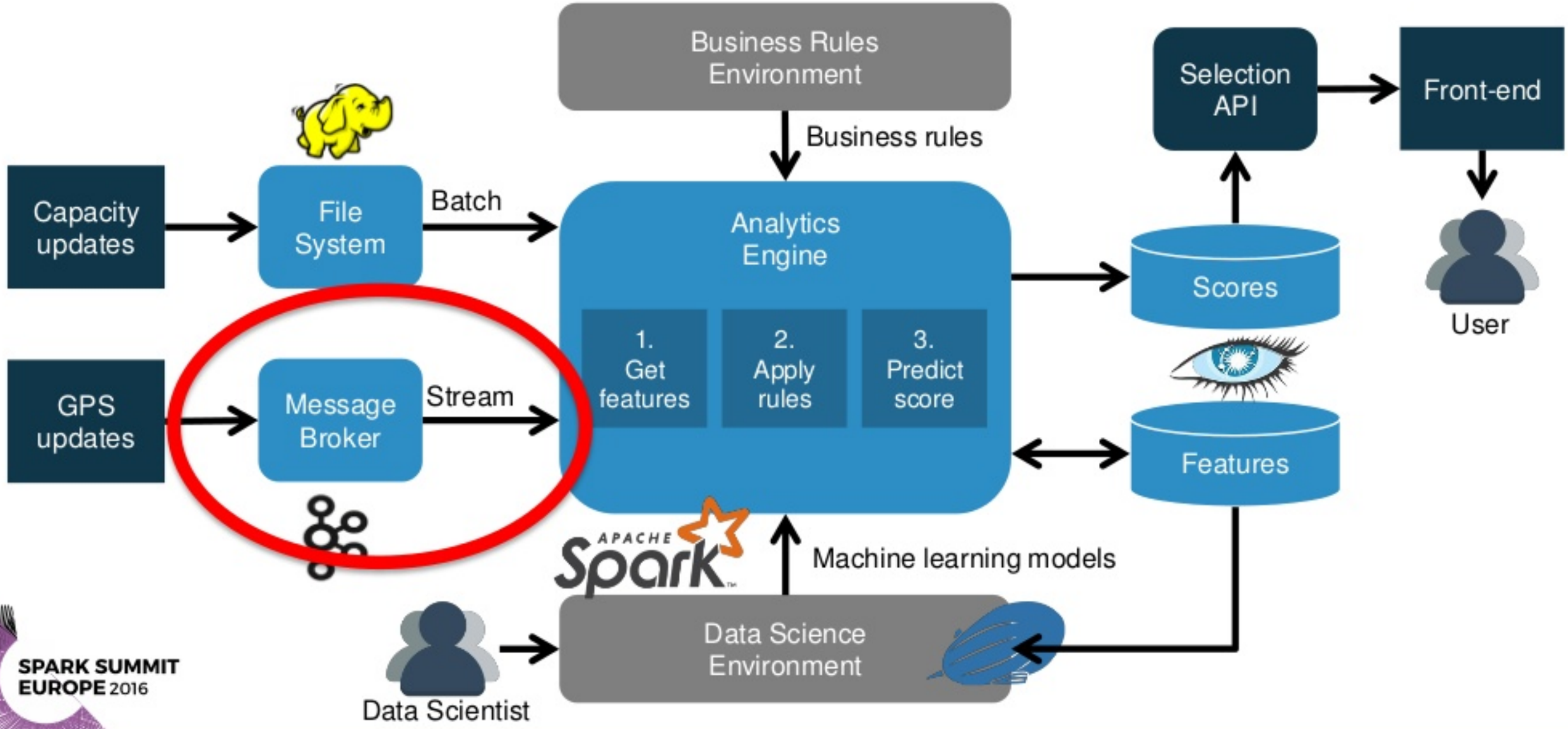
Lambda Architecture



Lambda Architecture



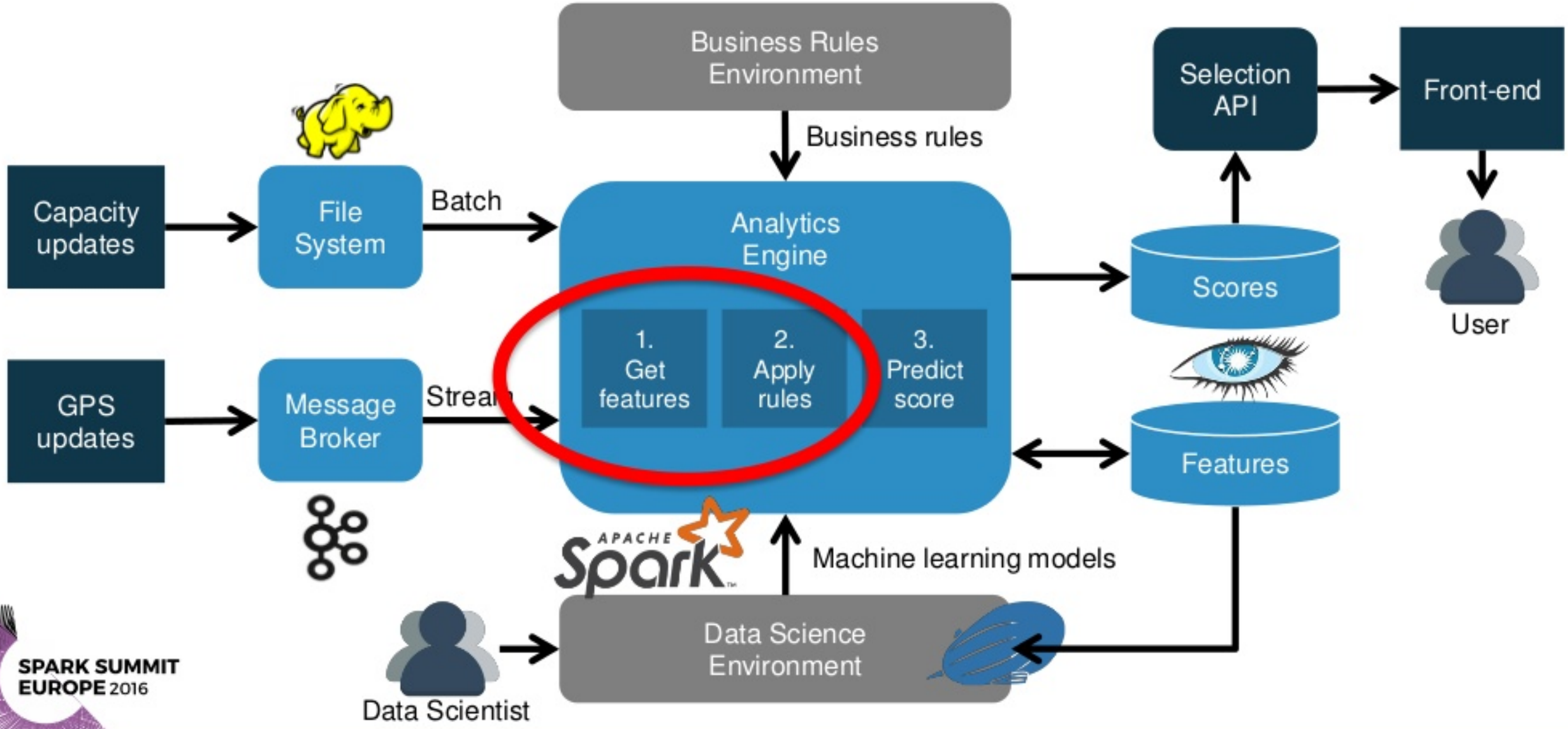
Lambda Architecture



Event Processing with Kafka

```scala
import org.apache.spark.streaming.{Seconds, StreamingContext}
import org.apache.spark.streaming.kafka010._
import org.apache.spark.streaming.kafka010.LocationStrategies.PreferConsistent
import org.apache.spark.streaming.kafka010.ConsumerStrategies.Subscribe

// initialize Spark Streaming
val conf = new SparkConf().setAppName("fast-data").setMaster("local[*]")
val ssc = new StreamingContext(conf, Seconds(1)) // batch interval = 1 sec

// set parameters for Kafka connection
val topics = Array("cars")
val kafkaParams = Map[String, Object]("bootstrap.servers" -> "localhost:9092")

// subscribe to stream -> create Spark DStream
val stream = KafkaUtils.createDirectStream[String, String](
  ssc,
  PreferConsistent,
  Subscribe[String, String](topics, kafkaParams))
```

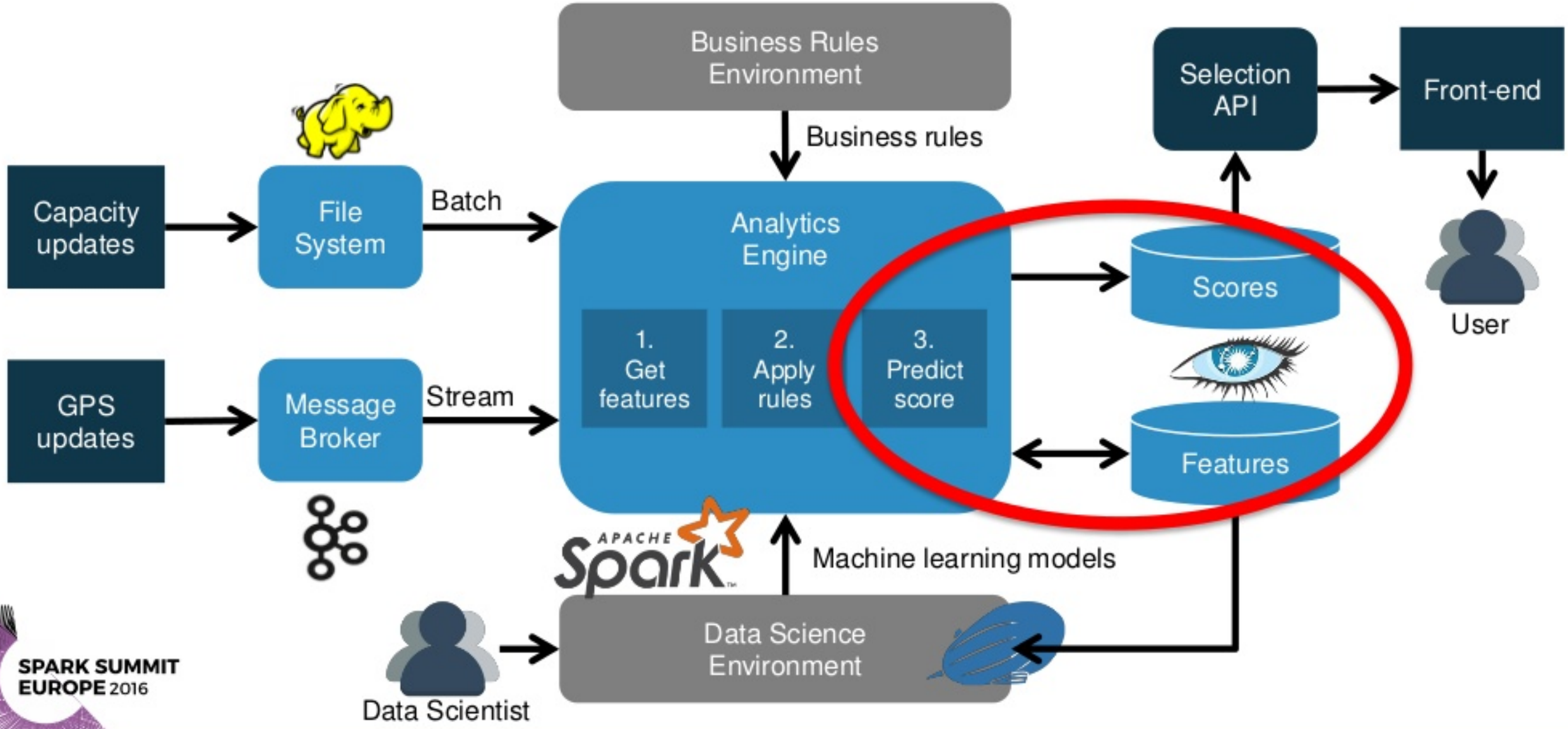
Lambda Architecture



Stream: Data Preparation

```
// change raw data to business events
val events = stream
  .map(event => CarLocationHelper.createCarLocation(event.value))

// apply business rules
val filtered = events
  .filter(Location.filterLocalArea)    // only select cars in local area

// store car locations (update or create)
filtered
  .foreachRDD(rdd => rdd.foreach(cle => { CassandraHelper.insertCarLocation(cle) }))
```

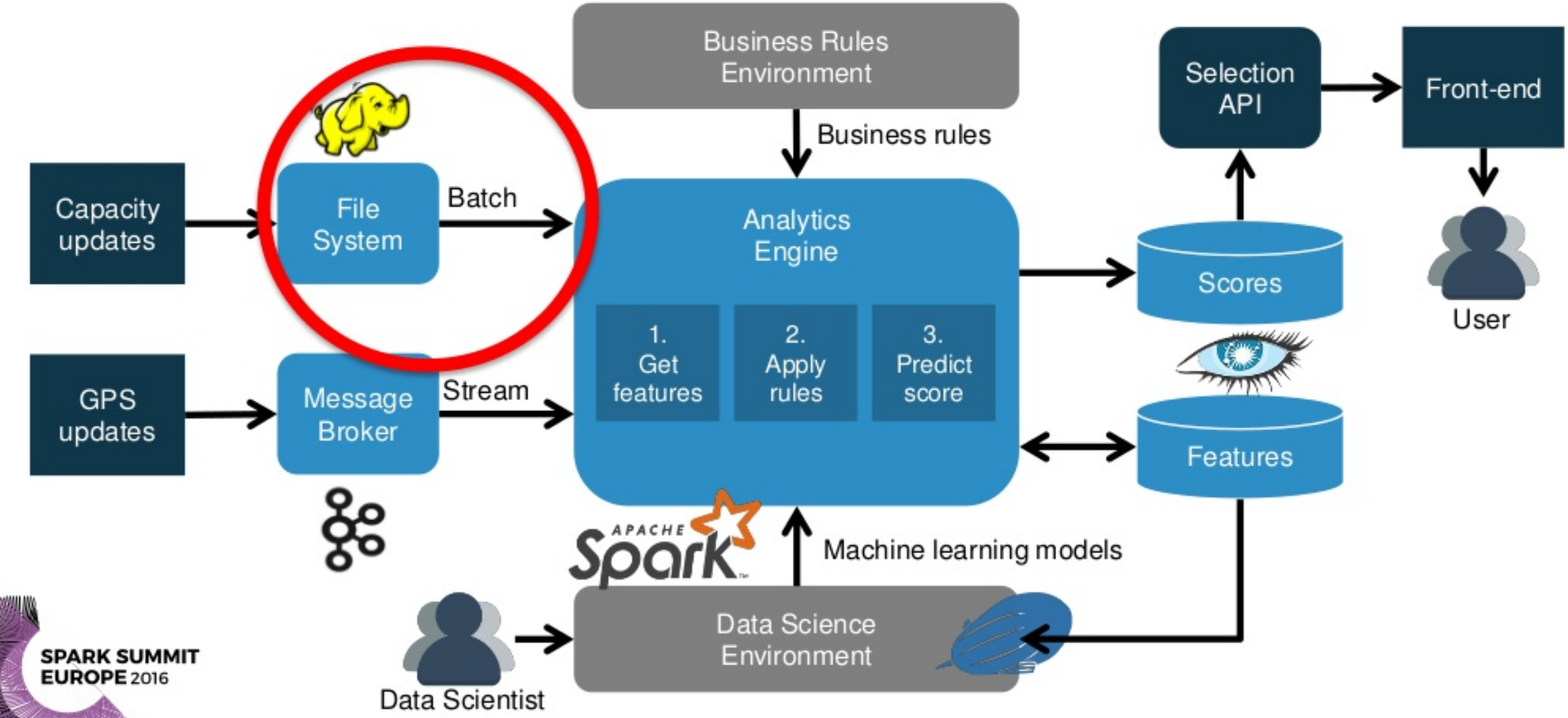
Lambda Architecture



Stream: Create Recommendation

```
// recalculate the distribution of each car park in the neighborhood
filtered
  // get feature set
  .map(cl => Location.getLocalCarParks(cl))
  .foreachRDD(rdd => rdd.foreach(carParks => carParks.foreach { cp =>
    // combine event with previous events (running average)
    val carsInNeighborhood = DensityCalculator.calculateDensity(cp, getCarsInNeighborhood(cp))
    val updatedCarPark = cp.setCarsInNeighborhood(carsInNeighborhood)

    // predict score (machine learning)
    val vector = new DenseVector(updatedCarPark.featureVectorArray)
    val score = model.predict(vector)

    // update feature set and score
    val scoredCarPark = updatedCarPark.setScore(score)
    updateCarParkFeatures(updatedCarPark)
  }
))
```

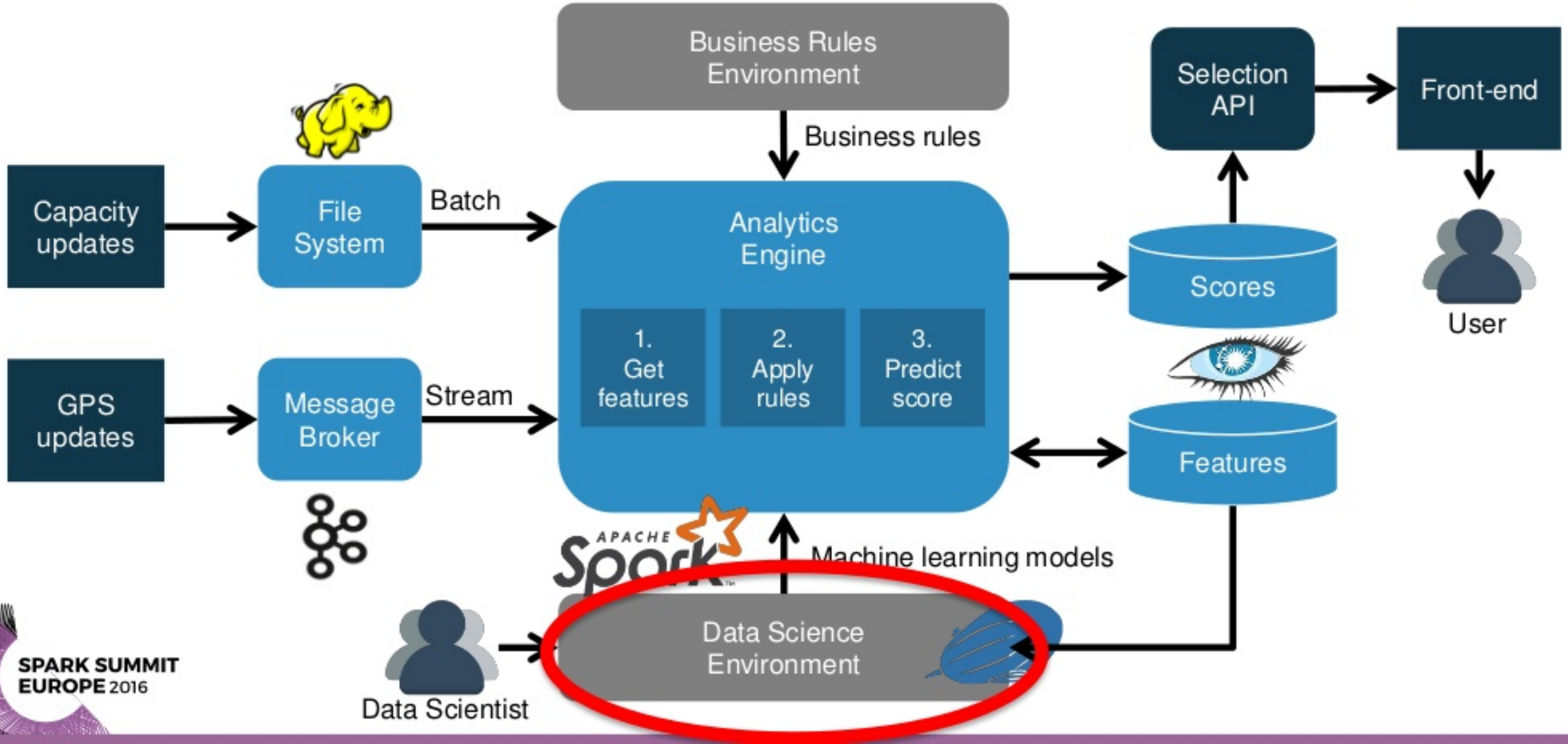
Lambda Architecture



Batch Processing

```
// initialize Spark for batch processing
val spark = SparkSession.builder.appName("batch-data").master("local[*]")
  .getOrCreate()

// clean up (remove car data older than 60 seconds)
CassandraHelper.removeOldCarLocations(60)

// get car park data
val textFile = spark.sparkContext.textFile("hdfs://data/latest.csv")

// iterate over all car parks in the data set
textFile.map(line => CarParkHelper.createCarPark(line)).map{cp =>
  // get recent set of car data (running average)
  val carsInNeighborhood = DensityCalculator.calculateDensity(cp, getCarsInNeighborhood(cp))
  val updatedCarPark = cp.setCarsInNeighborhood(carsInNeighborhood)

  // predict score (machine learning)
  val vector = new DenseVector(updatedCarPark.featureVectorArray)
  val score = model.predict(vector)

  // update feature set and score
  val scoredCarPark = updatedCarPark.setScore(score)
  updateCarParkFeatures(updatedCarPark)
}
```

Lambda Architecture



Data Science with Zeppelin

http://localhost:9090/

localhost:9090/#/notebook/2BZGHGJXH

Zeppelin Notebook

Search your Notebooks

anonymous

fast-data

default

```
import org.apache.spark.mllib.regression.LabeledPoint
import org.apache.spark.mllib.regression.LinearRegressionModel
import org.apache.spark.mllib.regression.LinearRegressionWithSGD

println("Car park recommendations")

case class CarPark(name: String, latitude: Float, longitude: Float, capacity: Int, usage: Float, accessibility: Float, openFrom: Int, openUntil: Int, rate: Float, cars: Float, score: Float)

// load data
var file = sc.textFile("/home/bas/data.csv")
val records = file
   .map(l => l.split(","))
   .map(parts => CarPark(parts(0), parts(1).toFloat, parts(2).toFloat, parts(3).toInt, parts(4).toFloat, parts(5).toFloat, parts(6).toInt, parts(7).toInt, parts(8).toFloat, 0, 0))

// prepare data, create vector
val data = records.map(record => LabeledPoint(record.score, Vectors.dense(record.cars, record.rate, record.usage))).collect()
val rdd = spark.sparkContext.makeRDD(data)

// separate training and test set
val splits = rdd.randomSplit(Array(0.8, 0.2))
val training = splits(0).cache()
val test = splits(1).cache()

// train the model
val algorithm = new LinearRegressionWithSGD()
algorithm.setIntercept(true)
algorithm.optimizer.setNumIterations(100).setStepSize(0.1)
val model = algorithm.run(training)

println(model.intercept)
println(model.weights)

// do a test run
val prediction = model.predict(test.map(_.features))

// check output
val predictionAndLabel = prediction.zip(test.map(_.label))
predictionAndLabel.count()
predictionAndLabel.take(20).foreach(p => println("Prediction=" + p._1 + ", actual=" + p._2))

model_health.save(sc, "/home/bas/models/health")
```

Summary

- The IoT requires new architecture principles: in-memory, distributed, reactive and scalable are the new normal
- Lambda/Kappa/Omega reference architectures are designed for combining batch and streaming data flows
- Open source tooling such as Kafka, Cassandra, and Spark adhere to these principles and design patterns

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